



Zinc supply influenced the growth, yield, zinc content, and expression of ZIP family transporters in sorghum

Theivanayagam Maharajan¹ · Thumadath Palayullaparambil Ajeesh Krishna¹ · Stanislaus Antony Ceasar¹ · Savarimuthu Ignacimuthu²

Received: 22 June 2022 / Accepted: 13 January 2023 / Published online: 23 January 2023
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract

Main conclusion Zinc deficiency altered shoot and root growth, plant biomass, yield, and ZIP family transporter gene expression in sorghum.

Abstract Zinc (Zn) deficiency affects several crop plants' growth and yield, including sorghum. We have evaluated the sorghum under various concentrations of Zn supply for phenotypic changes, Zn content, and expression of Zn-regulated, iron-regulated transporter-like proteins (ZIP) family genes. Zn deficiency reduced the shoot and root growth, plant biomass, and yield by > 50%. The length and number of lateral roots were increased by more than 50% under deficient Zn compared to sufficient Zn. Ten *SbZIP* family transporter genes showed dynamic expression in shoot and root tissues of sorghum under deficient and sufficient Zn. *SbZIP2*, 5, 6, 7, and 8 were expressed in all tissues under deficient and sufficient Zn. *SbZIP2*, 4, 5, 6, 7, 8, and 10 were highly induced in shoot tissues by deficient Zn. The expression level of *SbZIP6*, 7, 8, and 9 was higher in root tissues under deficient Zn. The phylogenetic analysis revealed that most *SbZIP* family proteins are closely associated with the *ZmZIP* family of maize. The functional residues His177 and Gly182 are fully conserved in all *SbZIP* family transporters, as revealed by homology modeling and multiple sequence alignment. This study may provide a foundation for improving the Zn-use efficiency of sorghum.

Keywords Gene expression · Genome-wide identification · Phenotyping · Phylogenetic analysis · Zn-regulated, iron-regulated transporter-like proteins (ZIP) · Zn deficiency

Abbreviations

MC	Main cluster
SDW	Shoot dry weight
RDW	Root dry weight
SL	Shoot length
PRL	Primary root length
SW	Seed weight

TMDs	Transmembrane domains
LRL	Lateral root length
LRN	Lateral root number
ZIP	Zn-regulated, iron-regulated transporter-like proteins

Communicated by Dorothea Bartels.

Theivanayagam Maharajan and Thumadath Palayullaparambil Ajeesh Krishna contributed equally.

✉ Stanislaus Antony Ceasar
saceasar@rajagiri.edu; antony_sm2003@yahoo.co.in

¹ Division of Plant Molecular Biology and Biotechnology, Department of Biosciences, Rajagiri College of Social Sciences, Kochi, Kerala 683104, India

² Xavier Research Foundation, St. Xavier's College, Palayamkottai 627002, India

Introduction

Sorghum (*Sorghum bicolor* L. Moench) is the fifth most important staple cereal crop for humans and animals, providing food, fodder, fiber, and fuel. Like rice (*Oryza sativa*), maize (*Zea mays*), and small millets, sorghum belongs to the family Poaceae and is a C₄ cereal crop. It is cultivated primarily in Asia and Africa's tropical and subtropical regions (Mundia et al. 2019). In these countries, sorghum shows comparative advantages over other summer cereals, including its capacity to fill grain during end-of-season droughts (Maharajan et al. 2021a). Sorghum is cultivated mainly

for animal feed in developing countries such as the United States, Europe, and Brazil (Mundia et al. 2019; Maharajan et al. 2021a). More than 35% of sorghum is used for human consumption, and the rest for animal feed and alcohol production (Gerrano et al. 2014). Like other cereals, sorghum is a good source of vitamins and minerals such as potassium (K), phosphorus (P), zinc (Zn), sodium (Na), calcium (Ca), and iron (Fe) (Abah et al. 2020; Maharajan et al. 2021b). It is considered a cereal that meets the nutritional needs of poor people for a healthy life and improves their livelihood. Globally, cereals are cultivated on about 700 million hectares of agricultural land. Sorghum is cultivated on over 40 million hectares of agricultural land worldwide (Mundia et al. 2019). The growth and development of sorghum are affected by biotic and abiotic stresses, including micronutrient deficiency (Maharajan et al. 2021a). Zn deficiency reduces plant height, primary root length (PRL), leaf width, dry weight of shoot and root (SDW and RDW), total plant biomass, and Zn content in leaf and root tissues of sorghum (Li et al. 2013a). Further research is needed to develop sorghum cultivars that can grow under low Zn soil, which will help improve sorghum production and alleviate malnutrition among poor people.

In general, both macro- and micro-nutrients are required at optimum concentrations for the growth and development of plants (Maharajan et al. 2022a). Zn is one of the essential micro-nutrients, and its low availability reduces plant growth and yield (Krishna et al. 2017). Zn plays an essential role in carbohydrate metabolism, maintaining the integrity of cellular membranes, protein synthesis, regulation of auxin synthesis, pollen formation, stabilization of ribosomal fractions, and synthesis of cytochrome (Krishna et al. 2020; Yuvaraj and Subramanian 2020). Zn is mainly acquired and transported in plants as a divalent (Zn^{2+}) cation (Gupta et al. 2016). Zn^{2+} has to pass through several tissues from the root (epidermis, exodermis, cortex, endodermis, and pericycle) before reaching the xylem for transport to aerial parts (Gupta et al. 2016). The transport mechanism of Zn^{2+} differs in monocots and dicots. In dicots, Zn^{2+} is transported across the epidermis, cortex, endodermis, and pericycle (Amini et al. 2022). In monocots, it is transported across the epidermis, exodermis, cortex, endodermis, and pericycle (Krishna et al. 2022a). The additional diffusion barrier of the exodermis is located above the cortex and below the endodermis in monocots (Gupta et al. 2016). After crossing the barrier of the Casparian strip in the root endodermis, root-to-shoot Zn^{2+} transport occurs via xylem loading and is then redistributed via the phloem cells (Clemens 2001). The monocots release phytosiderophores (PS) (such as nicotianamine and mugineic acids) in the rhizosphere, which chelates Zn and increases Zn availability (Amini et al. 2022). The Poaceae species respond to Zn deficiency by exuding PS to increase the availability and transport of the PS-Zn

complex, which has been reported in rice, barley (Suzuki et al. 2006), and maize (Von Wiren et al. 1996).

A special type of membrane or carrier protein is required for Zn^{2+} transport in plants. Various families of membrane transporters such as Zn-regulated, Fe-regulated transporter-like proteins (ZIP), yellow stripe-like proteins (YSLs), heavy metal ATPases (HMAs), metal tolerance proteins (MTPs), Zn-induced facilitators (ZIFs) and cation diffusion facilitator (CDF) are involved in both uptake and transport of Zn from the soil to various tissues in plants (Krishna et al. 2020; Maharajan et al. 2022b). Dicot plants uptake Zn^{2+} as free divalent cations from the soil, whereas monocot plants uptake both, divalent cation Zn^{2+} and chelated forms (Gupta et al. 2016). In particular, the chelation of Zn^{2+} by PS plays a significant role in Zn^{2+} uptake in monocots (Kolaj-Robin et al. 2015). Monocot plants uptake Zn-PS complexes through YS1 transporters (Nozoye et al. 2015).

Interestingly, the ZIP family transporter has been proven to be involved in the uptake of free Zn^{2+} under low Zn conditions in monocots (Huang et al. 2020). In rice, the ZIP family transporter (OsZIP9) has been proven to be involved in Zn^{2+} uptake under low Zn conditions (Huang et al. 2020). The role of other monocot ZIP family transporters involved in acquiring free Zn^{2+} has also been reported for many monocot plants (Krishna et al. 2020). Most of these genes were regulated by external Zn supply in monocots, and many genes were also induced by Zn deficiency in rice, maize, wheat, and barley (Amini et al. 2022). Therefore, we have analyzed the expression response of ZIP family genes in shoot and root tissues of sorghum under deficient and sufficient Zn conditions. ZIP family transporters help to carry Zn^{2+} ions across cellular membranes into the cytoplasm (Eide et al. 1996; Conte and Walker 2011). The ZIP family of transporters not only transports Zn^{2+} across membranes, but it can also transport other divalent metal cations such as Mn^{2+} , Co^{2+} , Cu^{2+} , Cd^{2+} , Fe^{2+} , and Ni^{2+} (Hall and Williams 2003). Several ZIP family transporters have previously been identified and characterized in many kinds of cereal. A total of 16, 14, 13, and 12 ZIP family transporters were identified in various tissues of rice (Ramesh et al. 2003; Ishimaru et al. 2005; Chen et al. 2008), wheat (*Triticum aestivum*) (Evens et al. 2017), barley (*Hordeum vulgare*) (Tiong et al. 2015) and maize (Mondal et al. 2014), respectively, under deficient and sufficient Zn conditions. Among the 16 OsZIPs of rice, OsZIP1 and 3 were reported to be responsible for Zn uptake from soil (Bashir et al. 2012). OsZIP4, 5, and 8 seem important for the translocation of Zn from root to shoot, and OsZIP4 and 8 play a key role in rice grain filling (Ishimaru et al. 2007; Lee et al. 2010). The ZmZIP2, 5, 6, 8, and 11 were involved in grain filling, because their expression levels were higher in kernel tissues of maize under deficient Zn conditions (Mondal et al. 2014). Four genes (*TaZIP3*, 5, 7, and 13) were

highly expressed in roots under deficient Zn conditions and involved in Zn uptake in wheat (Evens et al. 2017). Many ZIP family transporters were analyzed in root, shoot, leaf, culms, flag leaf, kernel, and spikelets of various other plants under deficient Zn conditions as reviewed by Krishna et al. (2020).

No ZIP family transporters have yet been identified and characterized in sorghum under deficient and sufficient Zn conditions. Paterson et al. (2009) have sequenced the whole-genome sequence of sorghum. It allows us to identify and analyze the ZIP family of genes and proteins in sorghum. We have identified 10 ZIP (SbZIP1 to SbZIP10) family transporters from the sorghum genome sequence. This study analyzed their expression profiles in shoot and root tissues under deficient and sufficient Zn conditions. The predicted protein sequences of SbZIP family transporters were analyzed for the presence of putative functional residues involved in the Zn transport through homology modeling and ClustalW alignment using transporters with available crystal structure as a template. The phylogenetic relationships of the SbZIP family with those of other plants were also analyzed. We have also grown the sorghum under eight different concentrations (0.05, 0.25, 0.5, 0.75, 1, 2, 5, and 10 μM) of Zn and identified the deficient, sufficient and toxicity levels of Zn conditions for sorghum growth and development based on the phenotypic traits, plant biomass, yield, and Zn content data. This study will help researchers know about the deficient, sufficient, and toxicity levels of Zn conditions for sorghum growth and development and may form the foundation for further studies to overcome the Zn deficiency in sorghum.

Materials and methods

Computational analysis

Identification of SbZIP family transporters in the genome of sorghum

The rice (*O. sativa*) and foxtail millet (*Setaria italica*) protein sequences of ZIP family transporters were used for “tblastn” search with sorghum genome assembly (*S. bicolor* V2.1 assembly) at the Phytozome website (<https://phytozome-next.jgi.doe.gov/>). A total of 10 SbZIP family transporters were collected from the genome assembly of sorghum (Table 1). We assigned gene numbers for SbZIP family transporters based on the sequence similarity of rice (Table 1). The selection was based on, except value (E) ($E=0$), the presence of start codon “ATG”, scaffolds, and sequencing range. Based on these components, we downloaded the genomic, CDs, and protein sequences of the ZIP family of transporters of sorghum. SbZIP family

transporter protein sequences were validated at the Uniprot database and are confirmed to be ZIP family transport proteins (Table 1). SbZIP family genes were mapped on the corresponding chromosomes of sorghum by the MapGene2chromosome web v2 (MG2C) database (http://mg2c.iask.in/mg2c_v2.0/) according to their position information, available on the Phytozome database (*S. bicolor* V2.1 assembly). Gene structure display server program (GSDS, <http://gsds.cbi.pku.edu.cn/>) was used to detect the intron/exon distribution in the genomic sequences of SbZIP transporters.

Primers for SbZIP family and constitutive (*SbEF4 α*) genes were designed by Primer 3 output version 14 software (<http://bioinfo.ut.ee/primer3-0.4.0/>). After primer designing, we verified the specificity of each primer using “blastn” tool on the Phytozome website at the sorghum genome assembly. Among the 10 SbZIP genes, we have designed primers for SbZIP2, 3, 4, 7, 8, 9, and 10 genes with intervening intron between the primers and for the other two genes (*SbZIP1* and 6) without introns for qRT-PCR experiments to detect genomic DNA contamination (Table 2).

Analysis of protein features of SbZIP family transporters

The number of transmembrane domains (TMDs) of each SbZIP transporter was predicted by the TMHMM server v.2.0, an online-enabled tool. Physicochemical properties such as molecular weight (MW) and isoelectric point (pI) of each SbZIP protein were determined by ProtParam tool on the ExPASy server. The subcellular localization of each SbZIP protein was predicted by the WoLF PSORT server (<https://www.genscript.com/wolf-psort.html>).

Phylogenetic analysis of SbZIP family transporters

A phylogenetic tree was constructed using the protein sequences of SbZIP family transporters of sorghum with those of other Poaceae family members such as foxtail millet and green foxtail (*Setaria viridis*), sorghum, maize, barley, wheat, rice, and of the model plant Arabidopsis. The tree was constructed by molecular evolutionary genetics analysis (MEGA) v6 software tool. Protein sequences of ZIP family transporters were first aligned separately using the ClustalW method, and the evolutionary tree was deduced based on the Jones–Taylor–Thornton matrix-based model, and phylogeny was then constructed via the maximum-likelihood (ML) method. The evolutionary analyses were conducted using the MEGA v6 tool. The bootstrap value of 1000 replicates was maintained to construct the phylogenetic tree. The phylogenetic tree was visualized and edited using FigtreeV1.4.2

Table 1 Details of zinc-regulated, iron-regulated transporter-like proteins (ZIP) family transporters of sorghum

S. No	Name of the gene	Gene locus	Location	Length of CDS sequence	Length of Protein sequence	Chromosome number	Number of trans membrane domains (TMDs)	Molecular weight (kDa)	Isoelectric point (pI)	Sub-cellular localization	UniProt ID
1	SbZIP1	Sobic.003G443700.1	74,163,178...74164808	1086	361	Chr03	9	38.86	8.74	Plasma membrane	Q94DG6_OsZIP1
2	SbZIP2	Sobic.001G339600.1	62,735,381...62738856	1077	358	Chr01	9	37.42	5.62	Plasma membrane	Q852F6_OsZIP2
3	SbZIP3	Sobic.006G212200.1	56,118,591...56121279	1104	367	Chr06	6	38.61	7.87	Plasma membrane	Q7XLD4_OsZIP3
4	SbZIP4	Sobic.007G075700.1	8,699,712...8702902	1209	402	Chr07	7	40.64	7.02	Plasma membrane	Q6Z191_OsZIP4
5	SbZIP5	Sobic.009G170500.1	52,613,311...52616367	1131	376	Chr09	7	38.43	5.97	Plasma membrane	Q6L8G0_OsZIP5
6	SbZIP6	Sobic.009G057700.1	5,972,474...5975450	1227	408	Chr09	8	42.72	5.95	Plasma membrane	Q6L8F9_OsZIP6
7	SbZIP7	Sobic.009G073100.1	9,075,915...9078371	1167	388	Chr09	6	40.12	6.15	Plasma membrane	Q6L8F7_OsZIP7
8	SbZIP8	Sobic.002G083200.2	8,889,038...8893535	1149	382	Chr02	7	39.42	6.33	Plasma membrane	A3B111_OsZIP8
9	SbZIP9	Sobic.007G064900.1	6,823,204...6826295	1092	363	Chr07	9	37.84	6.08	Plasma membrane	Q0DHE3_OsZIP9
10	SbZIP10	Sobic.010G171300.1	50,278,532...50284909	1329	442	Chr10	6	45.21	5.67	Plasma membrane	Q5Z653_OsZIP10

Table 2 Details of primers for the zinc-regulated, iron-regulated transporter-like proteins (ZIP) family of sorghum with annealing temperatures (T_m) used for qRT-PCR experiment

Name of the genes	Forward primer (5' to 3')	Reverse primer (3' to 5')	Product size with intron (bp)	Product size without intron	Temperature (°C)
<i>SbZIP1</i>	CCTCCTGGGCACACAGTTC	ACGGGTACTGGTCTTGGTG	–	110	60.7
<i>SbZIP2</i>	CATCTCCTACGTCGTCTCACG	AGGTGGTGTGGGATCAGAG	837	121	59.6
<i>SbZIP3</i>	GATCCATGAACAACCTGGAGAT	TGAATGCACCAAGATTCCAA	673	193	60.1
<i>SbZIP4</i>	GTTCTGGAGCTGGGAATCCT	TGCTGTAGCTGGAAGTGATG	896	252	59.0
<i>SbZIP5</i>	GACGTTCCACCAGCTCTTC	GACTGCCATCGCTAACATTG	534	91	58.8
<i>SbZIP6</i>	GGAATGCTAGTGTTCATATGAC	ACATAGAAGCAGATCCAAGCAC	–	217	58.0
<i>SbZIP7</i>	ACATGTCCGCTCCATCGT	AGAACTGGTGGAAATGCAAGC	233	143	60.1
<i>SbZIP8</i>	GCTTCCACCAAATGTTTCGAG	CACCATGGTCACGATCGAC	970	86	60.8
<i>SbZIP9</i>	CCATCGCGTCATCTCTCAG	GGACTCTCAGATGCGCCTAC	233	77	60.2
<i>SbZIP10</i>	TTGTGATGACGGTCACAATG	ACTAGAGGCTTGATGGTACAGG	950	156	58.2
<i>SbEF4α</i>	CTGGATGAAGCTGATGAGATG	GCATGGTAGCTGAGAACACAC	880	109	58.4

(<http://tree.bio.ed.ac.uk/software/figtree/>) tool. Then, Micro-soft PPT was used for assigning main clusters (MCs).

Prediction of functional residues in SbZIP protein sequences of sorghum by multiple sequence alignment

Conserved and functionally important amino acid residues of SbZIPs (SbZIP1–SbZIP10) of sorghum were predicted by multiple sequence alignment using ClustalW in MEGA v6.0 tool using *Bordetella bronchiseptica* ZIP (BbZIP) as a template. The truncated protein sequences SbZIPs (File S1) were aligned based on homology modeling (see below) with the protein sequences of BbZIP with the known crystal structure (Zhang et al. 2017) as a template to identify the conserved residues involved in Zn binding and transport.

Homology modeling of SbZIP proteins

Homology models were constructed for SbZIPs using BbZIP as a template, whose crystal structure had been solved. Models of 10 ZIP family transporters of sorghum (SbZIP1 to SbZIP10) were deduced using BbZIP (Zhang et al. 2017) (UniProt id: A0A0H3LM39; PDB id: 5TSA) as a template. Truncated sequences of SbZIP proteins were used for modeling due to the presence of extensions at N and C terminals and at internal sequences (File S1). Homology models were generated using Modeler v9.22, which created 100 models for each target protein. The models were then ranked based on their modeler objective function value. The five best models were analyzed using MolProbity (<http://molprobity.biochem.duke.edu/>), an online tool. Ramachandran plots were generated for these five models, and finally the best model was chosen based on the % residues in the

favorable and allowed regions and the number of outliers. The models were then visualized by LiteMol viewer, an online-enabled tool (<https://www.litemol.org/Viewer/>) and Pymol.

Phenotypic and physiological study

Plant growth for phenotypic traits, Zn content, and yield analyses

We have selected disease tolerant, high-yielding and medium-duration genotype (APK-1) for physiological, biochemical and molecular experiments in this study. This genotype has been predominantly cultivated in south India (particularly in Tamil Nadu) regions. Seeds of the APK-1 genotype of sorghum were obtained from Tamil Nadu Agricultural University, Coimbatore, India. Sorghum seeds were grown in nursery bags (25 × 33.5 cm; Vol. 5.5L) filled with horticultural grade perlite (Astrra Chemicals, Chennai, India). The experiment was conducted in the greenhouse at Rajagiri College of Social Sciences, Kochi, Kerala, India, with 27 ± 2 °C temperature, 16/8 h light and dark cycle, and 85% relative humidity (RH). The basal nutrient solutions were prepared based on the previously reported protocol (Ceasar et al. 2014; Maharajan et al. 2019) containing modified levels of Zn (0.05, 0.25, 0.50, 0.75, 1.0, 2.0, 5.0 and 10.0 μM). The pH of the nutrient solution was adjusted to 5.6–5.8 using 0.1 M H₂SO₄ or 0.1 M NaOH. Three replicates were maintained for each concentration of Zn. After sowing the seed, 500 mL of the respective nutrient solution was supplied to each nursery bag once in 3 days. The volume was equally increased for all bags as the seedlings grew larger. After seed germination, a total of 12–15 seedlings were maintained in each nursery bag for all concentrations of Zn. After 8 days, the seedlings were thinned to ten in each

nursery bag. After 28 day of growth, three seedlings in each nursery bag were used for phenotypic traits and Zn content analysis. We immediately stored the shoot (all aerial tissues including leaf) and root tissues at -80°C for Zn content analysis. The remaining seedlings were allowed to attain maturation, and the seed was set in each bag.

Phenotypic traits such as shoot length (SL), PRL, SDW, and RDW were analyzed in 28-day-old seedlings of sorghum grown under eight different (0.05, 0.25, 0.5, 0.75, 1, 2, 5, and 10 μM) levels of Zn supply. For the measurement of PRL, the roots were carefully detached from perlite granules without damaging the root system and washed to remove any adhering perlite particles using deionized water and blotted with tissue paper; then, the PRL was measured using a stainless-steel ruler. The SL was measured as the distance between the collar and the shoot tip. The shoot and root samples were dried in a 60°C hot air oven for 48 h, and dry weights were recorded. After the maturation of seeds, three plants from each nursery bag were taken for seed weight (SW) analysis. Each data set was recorded from three plants from each nursery bag, and the mean value was calculated ($n=9$).

Plant growth for gene expression and root phenotyping experiments

A separate plant growth experiment was conducted for *SbZIP* family gene expression and lateral root analysis in sorghum. Based on the phenotypic traits, Zn content, and yield data, we have chosen deficient (0.05 μM) and sufficient (2.0 μM) Zn conditions for plant growth. Sorghum plants were grown hydroponically under basal nutritional solutions (Ceasar et al. 2014; Maharajan et al. 2019) containing two different levels of Zn, such as deficient (0.05 μM) and sufficient (2.0 μM) Zn (Fig. 1a). The total capacity of the hydroponic tank was 12.5 L, and 10 L nutrient solution was maintained in each tank for plant growth. Seeds were initially grown in nursery bags filled with perlite and fed with deionized water free of nutrients for 1 week. The healthy seedlings (1-week-old) were transferred onto hydroponic tanks filled with the nutrient solution (containing sufficient Zn) for 7 days. After 7 days, the plants were treated with deficient and sufficient Zn conditions for 14 days. Each hydroponic tank had an equal number (6) of plantlets at the same time in each. Three replicates were maintained for each level of (deficient and sufficient) Zn. The nutrient solution was replaced every 2 days. For measurement of LRL and LRN, the entire root system was carefully detached from the hydroponic tank (under deficient and sufficient Zn conditions separately) and captured by Canon camera (Canon EOS 1500D 24.1 Digital SLR Camera with EF S18-55 is II Lens). After capturing the root system, LRL was analyzed by the ImageJ tool. As reported before, LRL was counted along

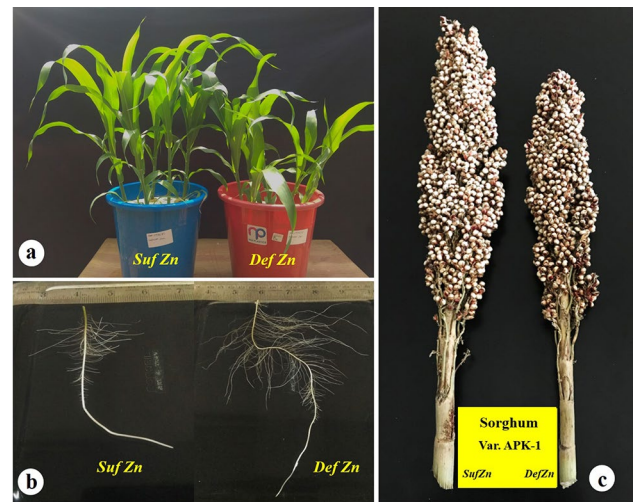


Fig. 1 Effect of Zn on the growth of seedlings, lateral root, and seed weight of sorghum. The sorghum plants are grown in a hydroponics system under deficient (0.05 μM) and sufficient (2 μM) Zn conditions (a). Lateral root images of sorghum seedlings grown under deficient (0.05 μM) and sufficient (2 μM) Zn conditions (b). Flower heads of sorghum grown under deficient (0.05 μM) and sufficient (2 μM) Zn conditions (c). The labeled Def Zn and Suf Zn represent deficient and sufficient Zn, respectively

a 5 cm length of the primary root (Maharajan et al. 2019). Three plants were collected from each hydroponics ($n=9$) tank and used to analyze the length and number of lateral roots under deficient and sufficient Zn conditions.

Analysis of Zn content

Zn content was assayed by a modified protocol of Liu et al. (1993) as described by Macnair and Smirnoff (1999). Zn contents were analyzed in the shoot (all aerial tissues including leaf) and root tissues of 28-day-old seedlings of sorghum grown under deficient (0.05 μM) and sufficient (2 μM) Zn conditions. A small quantity (5–40 mg) of the shoot and root tissues was collected separately in Eppendorf tubes (2 mL). Eppendorf tubes were immersed in liquid nitrogen (LN_2) for 30 min to freeze the shoot and root tissues. After 30 min, frozen tissues were ground separately with 500 μL 2% sulphosalicylic acid. Then, 20 μL of ground samples (extractant) were mixed with 780 μL of buffer (7.5 g KCl and 6.2 g boric acid in 1 L, pH adjusted to 9.6). After that, 200 μL of 0.03% Zincon (HiMedia, Mumbai, India) was added to the sample (20 μL extractant + 780 μL of buffer), and the absorbance was read at 606 nm by UV–Visible-spectrophotometer (Shimadzu, Model-UV1800). The Zn contents of the samples were calculated as per Macnair and Smirnoff (1999) method. A total of three replicates were maintained for analysis

of Zn content in shoot and root tissues of sorghum grown under both deficient and sufficient Zn. The Zn content was analyzed in the shoot and root tissues of three plants (from each nursery bag), and the mean value was calculated ($n=9$).

Molecular studies

RNA isolation and cDNA synthesis

For RNA isolation, shoot (all aerial tissues including leaf) and root tissues of 28-day-old seedlings were harvested separately from each hydroponic tank at the same time (10.30 to 11.00 am). Total RNA was isolated in triplicate for each sample using the NucleoSpin® RNA Plant kit (Machery-Nagel, Düren, Germany) according to the manufacturer's instructions. The purity and concentration of RNA were measured using a Nanodrop-spectrophotometer (ND-2000, Thermo Scientific, Waltham, MA, USA). cDNA was synthesized (in triplicate) from 500 ng of each total RNA using the QuantiTect Reverse transcription kit (Qiagen, Hilden, Germany). Six hydroponics tanks (three each for deficient and sufficient conditions) were maintained to grow the plant for RNA isolation. Three biological replicates were used for each sample for RNA isolation and cDNA synthesis.

Quantitative real-time RT-PCR (qRT-PCR)

Totally 10 *SbZIP* family genes (*SbZIP1* to *10*) and one constitutive gene (*SbEF4α*) of sorghum were used for qRT-PCR analysis (Table 2). Expression levels of 10 *SbZIP* family genes were analyzed in shoot and root tissues of 28-day-old seedlings of sorghum grown under deficient and sufficient Zn conditions. For quantification, 10 µL reactions were employed containing 5 µL of 2× SsoFast EvaGreen Supermix (Bio-Rad Laboratories, Hercules, CA, USA), 1 µL F:R primer mix (500 nM each primer), and 4 µL cDNA (1:50 dilution). The cycling conditions for qPCR were: enzyme activation at 95 °C for 30 s, denaturation at 95 °C for 5 s, annealing, and extension at 60 °C for 5 s (35 cycles). Finally, melting curve analysis was performed by increasing the temperature from 65 to 95 °C with 0.5 °C increments, followed by signal capture. All samples' cycle threshold (Ct) values were analyzed using the formula $2^{-\Delta\Delta Ct}$ (Roch et al. 2020). The Ct values were normalized using the values of *SbEF4α*. We have selected the constitutive gene (*SbEF4α*) for qRT-PCR data normalization based on the previous reports (Reddy et al. 2016; Wang et al. 2019). It has been proven to be the most stable gene for sorghum under various stresses. Three biological replicates and each consisting of three technical replicates were used for qRT-PCR analysis.

Statistical analysis

One-way ANOVA was used to determine significant differences among means. The data were analyzed using SPSS 17.0, and the values were considered significant at $P < 0.05$.

Results

Computational analysis

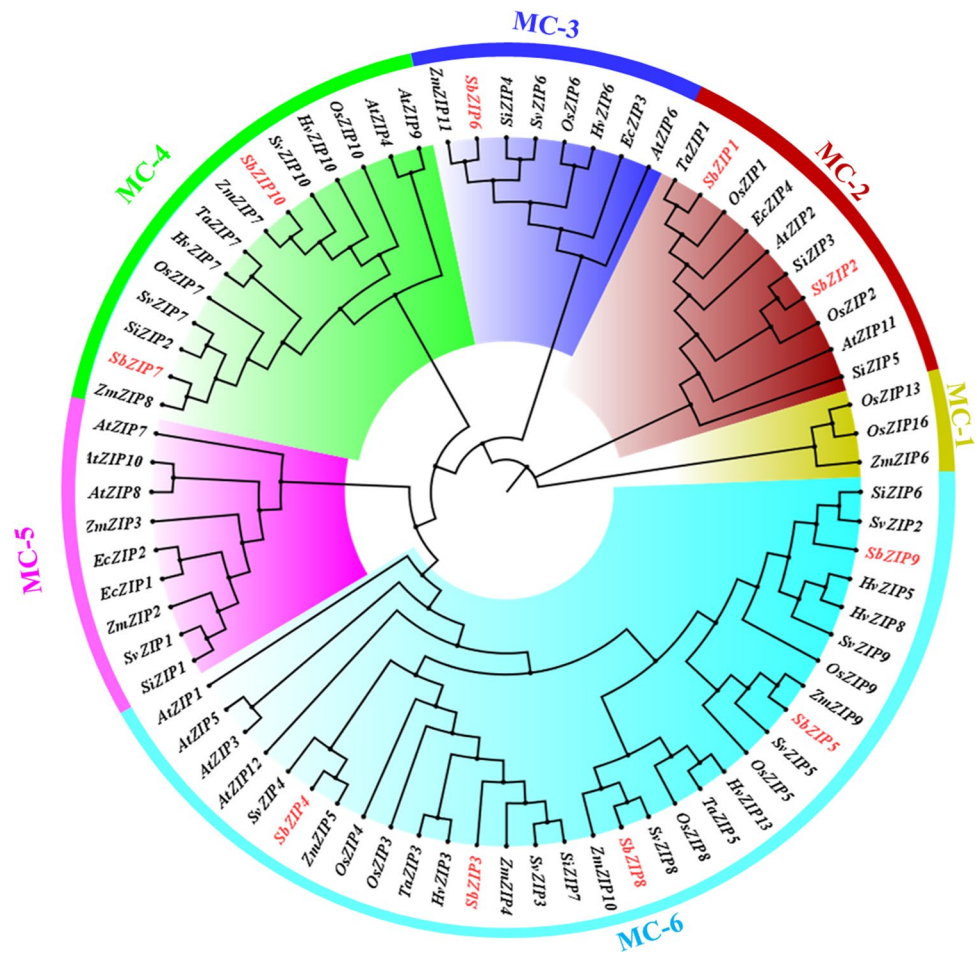
Protein features of ZIP transporters of sorghum

To identify the ZIP family members in sorghum, we used the protein sequences of rice OsZIPs and foxtail millet SiZIPs to perform a tblastn search against the genome assembly of sorghum (*S. bicolor* V2.1 assembly). A total of 10 *SbZIP* family gene sequences were retrieved from the Phytozome database after removing redundant sequences. The details of all 10 *SbZIP* genes, including genomic location, length of nucleotide and protein sequences, chromosome number, MW, pI, number of TMDs, and subcellular localization, are listed in Table 1. The tblastn analysis against the UniProt database showed that all of them belong to the ZIP family transporters (Table 1). The nucleotide lengths of the 10 *SbZIP* family genes ranged from 1,077 (*SbZIP2*) to 1,329 bp (*SbZIP10*). The length of *SbZIP* proteins ranged from 358 (*SbZIP2*) to 442 aa (*SbZIP10*). The predicted MW of *SbZIP* proteins ranged from 37.42 (*SbZIP2*) to 45.21 (*SbZIP10*) kDa, and the pI ranged between 5.62 (*SbZIP2*) and 8.74 (*SbZIP1*). All *SbZIP* proteins possess 6–9 TMDs (Table 1; Fig. S1) and are predicted to be localized to the plasma membrane (Table 1). The collected protein sequences of all 10 *SbZIP* family members are included in File S2. The mapmaker determined the chromosomal distributions of *SbZIP* genes, and they are distributed on 10 chromosomes (Fig. S2). Three *SbZIP* genes (*SbZIP5*, 6, and 7) are found on chromosome 09. Chromosome 7 contained two *SbZIPs* (*SbZIP4* and 9), while chromosomes 1, 2, 3, 4, 6, and 10 contained one *SbZIP* each (Fig. S2). The number of introns in the *SbZIP* family of genes varies from 0 to 4 (Fig. S3). The *SbZIP1* gene has no intron. *SbZIP6* has only one intron. The *SbZIP3*, 4, 5, 8, and 9 have two introns each; the *SbZIP2* and 7 have three introns each, and the *SbZIP10* has four introns (Fig. S3).

Phylogenetic relationship of SbZIPs

A phylogenetic tree was constructed with 76 plant ZIP protein sequences, including 10 *SbZIPs* of sorghum. The

Fig. 2 Phylogenetic analysis of sorghum SbZIP1 to 10 proteins. A phylogenetic tree of the sorghum ZIP family proteins was constructed with those of other Poaceae family members such as rice, maize, wheat, barley, sorghum, foxtail millet, green foxtail, and model plant *Arabidopsis*. The tree was constructed using 76 potential ZIP family transporter proteins from 9 plant species, including 10 sorghum SbZIP proteins, by MEGA version 6 software using the maximum-likelihood method based on the Jones-Taylor-Thornton matrix-based model with 1000 bootstrap replicates. The phylogenetic tree was visualized and edited using the FigtreeV1.4.2 tool. Then Microsoft PPT was used for labeling the main clusters (MCs). The phylogenetic tree was divided into six MCs and which are highlighted in gold (MC-1), brown (MC-2), blue (MC-3), green (MC-4), pink (MC-5), and aqua (MC-6). The SbZIP family proteins of sorghum are indicated in red



phylogenetic tree is comprised of 6 MCs, and each MC is subdivided into 1–3 sub-clusters (SCs) (Fig. 2). More ZIP proteins are clustered in MC-6 consisting of 32 ZIP transporters. It is followed by MC-4 (14 ZIPs) and MC-2 (10 ZIPs). The sorghum SbZIP1 is closely clustered with TaZIP1 and distantly clustered with OsZIP1 in the MC-2. In the same MC-2, SbZIP2 is closely and distantly associated with SiZIP3 and OsZIP2, respectively. Five transporters (SbZIP3, 4, 5, 8, and 9) are found in MC-6 and are closely and distantly clustered with monocot plant ZIPs (Fig. 2). For instance, SbZIP3 is distantly clustered with SvZIP3, ZmZIP4, and SiZIP7. SbZIP4, 5, and 8 are closely associated with ZmZIP5, ZmZIP9, and SvZIP8 in the MC-6. SbZIP9 (MC-6) is distantly clustered with SvZIP2 and SiZIP6. SbZIP6 is placed in MC-3 and closely clustered with the ZmZIP11. The remaining two proteins (SbZIP7 and 10) are found in MC-4. SbZIP7 is closely clustered with ZmZIP8, and SbZIP10 is closely clustered with ZmZIP7.

Prediction of functional residues of SbZIPs by multiple sequence alignment

We have identified the key functional residues of SbZIPs of sorghum at the metal-binding site using BbZIP as a template (Fig. 3). Most of the key functional residues of BbZIP are found in all SbZIP proteins of sorghum. Asp144 is involved in metal release in BbZIP and is conserved in the SbZIP7 transporter. His177 and Gly182 are involved in the metal release from the metal-binding site. These (His177 and Gly182) residues are fully conserved in all SbZIP family transporters (Fig. 3; Table 3). Leu200, Glu211, and Gly212 are essential for metal-binding in BbZIP, and these are conserved in all SbZIPs of sorghum except for SbZIP1 and 2.

Homology modeling

Homology models were produced for SbZIP proteins using BbZIP as a template with a known crystal structure to predict the key functional residues involved in Zn^{2+} binding and transport. The model of BbZIP revealed that Zn^{2+} binding is coordinated by the Glu181, Glu211, and Gln207 (Table 4; Fig. 4). However, the sorghum SbZIP proteins showed

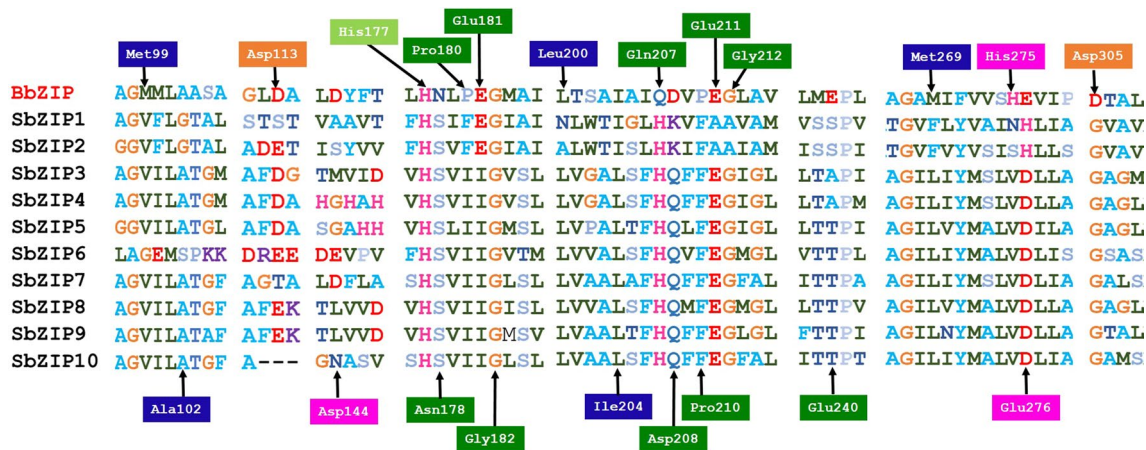


Fig. 3 Multiple sequence alignment of sorghum zinc-regulated, iron-regulated transporter-like proteins (ZIP) family transporter. The protein sequences of *Bordetella bronchiseptica* (BbZIP) and 10 SbZIPs of sorghum were aligned by ClustalW alignment using the molecular evolutionary genetics analysis, V 6.0 (MEGA6) tool. Asn178, Pro180, Glu181, Gly182, Gln207, Asp208, Pro210, Glu211, Gly212, and Glu240 are involved in metal binding and transport (highlighted in green). Hydrophobic residues of Met99, Ala102, Leu200, Ile204,

and Met269 are involved in blocking metals at the extracellular surface indicated in violet. Metal recruiting residues of Asp113 and Asp305 are highlighted in orange. Three residues (Asp144, His275, and Glu276) are involved in the metal release into the cytoplasm and are highlighted in pink. The His177 highlighted in light green is involved in both recruiting and cytoplasmic transport of metal

variation for residues available at the Zn^{2+} binding compared to BbZIP. Glu (Glu 181) is the only residue conserved at the Zn^{2+} binding of all the SbZIP proteins (Table 4; Fig. 4). SbZIP7 is the only protein having both Glu181 and Glu211 at the ligand-binding site. However, the positioning of the Glu residues also varied among SbZIP proteins due to the use of truncated sequences (File S1). Since the full-length sequence did not yield good models due to the presence of extension at N and C terminals and internal sequences (results not shown). In addition to Zn^{2+} coordinating residues, BbZIP showed Met 99, His 177, Met 269, and Glu 276 at the ligand-binding site (Table 4). These residues are highly diversified among SbZIPs. Other residues are not conserved besides His177, which is most prevalent among the SbZIPs. Met 269 of BbZIP is found only in 5 SbZIP proteins, and many novel residues like Arg, Leu, Ala, and Ser were found in SbZIPs at the ligand-binding site (Table 4).

Phenotypic and physiological study

Effect of Zn on growth and yield of sorghum

A total of four traits such as SL, PRL, SDW, and RDW were collected from 28-day-old seedlings of sorghum grown under eight different (0.05, 0.25, 0.50, 0.75, 1, 2, 5, and 10 μM) concentrations of Zn. The highest (10.83 cm) and lowest (3.4 cm) SL were observed at 2 and 0.05 μM Zn conditions, respectively (Fig. 5a). Under low Zn (0.05 μM), SL showed a 68.6% reduction compared to the other concentrations of Zn. The extremely higher concentration of Zn (10 μM) also

reduced the SL. At 10 μM Zn supply, SL was reduced by 60% compared to the 2 μM concentration of Zn (Fig. 5a). Zn supply also influenced the PRL of sorghum. The highest PRL (20.03 cm) was observed at 2 μM Zn. The mean values of PRL (< 12 cm) were observed at 0.05, 5, and 10 μM Zn (Fig. 5a). The reduction of PRL under low Zn (0.05 μM) by 51.5% was observed when compared to the 2 μM Zn. Higher concentrations of Zn (10 and 5 μM) reduced the PRL by 43.2 and 41.4%, respectively. However, the PRL increased under deficiency Zn on hydroponics growth (see below). The SDW and RDW were measured in shoot and root tissues of sorghum grown under various levels of Zn. Both SDW and RDW were reduced in sorghum grown under low Zn (0.05 μM). The highest SDW (420.2 mg) was obtained at 2 μM Zn (Fig. 5b). The SDW under 10 μM was 95.3 mg; at 0.05 μM it was 89.43 mg. These two levels of Zn reduced the SDW by > 75% compared to the other concentrations of Zn. As with SDW, the highest RDW was also obtained at 2 μM Zn. The mean RDW ranged from 34.0 (0.05 μM) to 208.9 mg (2 μM) (Fig. 5b). The RDW showed a significant reduction under low (0.05 μM) compared to high (2 μM) Zn. The reduction of RDW under 0.05 μM Zn was 83.7%. Sorghum grown under higher levels of Zn (5 to 10 μM) produced significantly lower mean values for RDW (Fig. 5b). These results indicated that lower and higher concentrations of Zn severely affected the growth and development of shoot and root tissues in sorghum.

The SW of sorghum was assessed after the maturation of seeds in sorghum under various concentrations of Zn. The length, width, and seed quality of the flowers were reduced by

Table 3 Conservation and positioning of functional residues in zinc and iron-regulated transporter proteins (SbZIP) family of sorghum aligned with *Bordetella bronchiseptica* zinc and iron-regulated transporter proteins (BbZIP)

Name of the protein	Name and position of the functional residues of BbZIP																			
	Met99	Ala102	Asp113	Asp144	His177	Asn178	Glu181	Gly182	Leu200	Ile204	Gln207	Asp208	Pro210	Glu211	Gly212	Glu240	Met269	His275	Glu276	Asp305
SbZIP1	Phe43	Thr46	Ser57	Ala89	His156	Ser157	Glu160	Gly161	Asn176	Ile180	His183	Lys184	Phe186	Ala187	Ala188	Ser218	Phe252	Asn258	His259	Gly287
SbZIP2	Phe99	Thr102	Glu113	Ser146	His216	Ser217	Glu220	Gly221	Ala236	Ile240	His243	Lys244	Phe246	Ala247	Ala248	Ser278	Phe312	Ser318	His319	Gly346
SbZIP3	Ile91	Thr94	Asp105	Met137	His157	Ser158	Ile161	Gly162	Leu177	Leu181	His184	Gln185	Phe187	Glu188	Gly189	Ala217	Leu253	Val259	Asp260	Gly288
SbZIP4	Ile60	Thr63	Asp74	Gly106	His152	Ser153	Ile156	Gly157	Leu172	Leu176	His179	Gln180	Phe182	Glu183	Gly184	Leu212	Leu248	Val254	Asp255	Gly283
SbZIP5	Ile98	Thr101	Asp112	Gly144	His186	Ser187	Ile190	Gly191	Leu206	Leu210	His213	Gln214	Phe216	Glu217	Gly218	Thr246	Leu282	Val288	Asp289	Gly317
SbZIP6	Glu73	Pro76	Glu87	Glu120	His168	Ser169	Ile172	Gly173	Leu188	Leu192	His195	Gln196	Phe198	Glu199	Gly200	Thr228	Leu266	Val272	Asp273	Gly302
SbZIP7	Ile96	Thr99	Thr110	Asp143	His192	Ser193	Ile196	Gly197	Leu212	Leu216	His219	Gln220	Phe222	Glu223	Gly224	Thr252	Leu288	Val294	Asp295	Gly323
SbZIP8	Ile93	Thr96	Glu107	Leu138	His180	Ser181	Ile184	Gly185	Leu200	Leu204	His207	Gln208	Phe210	Glu211	Gly212	Thr240	Leu276	Val282	Asp283	Gly311
SbZIP9	Ile91	Thr94	Glu105	Leu135	His172	Ser173	Ile176	Gly177	Leu192	Leu196	His199	Gln200	Phe202	Glu203	Gly204	Thr232	Leu268	Val274	Asp275	Gly303
SbZIP10	Ile86	Thr89	–	Asn129	His170	Ser171	Ile174	Gly175	Leu190	Leu194	His197	Gln198	Phe200	Glu201	Gly202	Thr230	Leu266	Val272	Asp273	Gly301

low Zn (0.05 μM) (Fig. 1c). Low (0.05 μM) Zn also delayed the time to flowering in sorghum by more than 15 days compared to 2 μM Zn supply (data not shown). The maximum mean SW (11.46 gm) was observed in sorghum grown under 2 μM Zn (Fig. 5c). The lowest mean SW (1.16 and 1.93 mg) were obtained in sorghum grown under extremely low (0.05 μM) and higher (10 μM) concentrations of Zn, respectively (Fig. 5c). The SW showed an 89.8% reduction under 0.05 μM Zn and 83% under 10 μM Zn supply compared to the value under 2 μM Zn supply. These results reveal that sorghum yield was severely affected by extremely low and higher concentrations of Zn. Both extremely low and higher concentrations of Zn also reduced the growth and development of shoot and root tissues. Based on these analyses of phenotypic traits, plant biomass, and yield data, we have found that a 2 μM concentration of Zn is sufficient for sorghum growth, development, and yield. We have also found that 0.05 μM is deficient, and > 5 μM are toxicity levels of Zn for growth, development, and yield of sorghum. So based on this study, we conducted further experiments such as Zn content, lateral root, and gene expression analysis only under deficient (0.05 μM) and sufficient (2 μM) concentrations of Zn.

We conducted a separate plant growth experiment for lateral root analysis under hydroponics. The sorghum seedlings were grown under deficient (0.05 μM) and sufficient (2 μM) Zn supply. We observed that leaf length and width were reduced under deficient Zn compared to sufficient Zn on hydroponics growth (Fig. 1a). Plants produced only 3 to 4 leaves under deficient Zn (0.05 μM) and had more than ten leaves under sufficient Zn (Fig. 1a). The PRL also showed an opposite effect on hydroponics growth. PRL was increased under deficiency Zn supply compared to the optimum Zn supply (Fig. 1b). The LRL and LRN were analyzed in 28-day-old seedlings grown under deficient and sufficient Zn conditions. The entire root system of seedlings grown under both conditions of Zn is presented (Fig. 1b). Sorghum grown under deficient Zn showed significantly higher mean LRL and LRN than those grown under sufficient Zn (Fig. 6a). Under sufficient Zn supply, the mean LRL and LRN were 2.8 and 16.3 cm, respectively. The LRL and LRN were 6.3 and 33.3 cm under deficient Zn supply. LRL and LRN production increased by 55.5 and 51%, respectively, under deficient Zn compared to sufficient Zn.

Deficient Zn influenced the Zn contents of the shoot and root tissues in sorghum

The Zn contents were measured in shoot and root tissues of sorghum grown under deficient (0.05 μM) and sufficient (2 μM) Zn supply. The root tissue showed higher Zn contents than shoot tissues under both deficient and sufficient Zn conditions (Fig. 6b). The root tissue had 1.12 $\mu\text{mol g}^{-1}$ FW (deficient Zn supply) and 2.25 $\mu\text{mol g}^{-1}$ FW (sufficient Zn supply). Shoot samples contained significantly higher Zn

Table 4 Residues predicted to be involved in Zn²⁺ binding through homology modeling using BbZIP as a template

Name of the protein	UniProt Accession Id	Length (amino acids)	Number of TMDs	Ligand (Zn ²⁺) coordinating residues	Ligand binding site residues
BbZIP (template)	BB2405	309	8	Glu181; Glu211 Gln207	Met 99; His177; Met 269; Glu276
SbZIP1	Sobic.003G443700.1	361	9	Glu160	Arg 26; Leu 52; His183; Ala 256;
SbZIP2	Sobic.001G339600.1	358	9	Glu 156	Pro 61; Met 65; Phe 182; Ser 252; Leu 256
SbZIP3	Sobic.006G212200.1	367	6	Glu 188	Ala 87; Ile 91; Ile 161; His 184
SbZIP4	Sobic.007G075700.1	402	7	Glu 183	Ala 56; Ile 60; Ile 156; His 179
SbZIP5	Sobic.009G170500.1	376	7	Glu 217; Ser 34	Arg 42; Ala 94; Ile 98; Ile 190
SbZIP6	Sobic.009G057700.1	408	8	Glu 199	Ser 75; His 168; His 195; Met 269
SbZIP7	Sobic.009G073100.1	388	6	Glu 186; Glu 223	Gly 67; Val 71; His 219; Met 291
SbZIP8	Sobic.002G083200.2	382	7	Glu 211	Phe 83; Ile 93; Ile 184; His 207;
SbZIP9	Sobic.007G064900.1	363	9	Glu 203	Ala 87; Ile 91; His 199; Asn 288
SbZIP10	Sobic.010G171300.1	442	6	Glu 201	Ala 82; Ile 86; His 197; Met 269

Name of the protein, UniPort Id, length of the protein, residues involved in ligand (Zn²⁺) binding are listed

TMDs transmembrane domain

(1.68 $\mu\text{mol g}^{-1}$ FW) under sufficient Zn than deficient Zn (Fig. 6b). Zn contents were reduced by 39.2 and 50.2% in shoot and root tissues of sorghum, respectively, under deficient Zn compared to the sufficient Zn.

Molecular studies

Expression analysis of *SbZIP* family genes in shoot and root tissues of sorghum grown under deficient and sufficient Zn conditions

qRT-PCR analysis was performed to examine the expression profiles of *SbZIP* genes in shoot and root tissues of sorghum grown under deficient and sufficient Zn supply. The *SbEF4 α* was used as a reference gene for the normalization of expression levels of *SbZIP* genes. The melting curve images of all the samples produced a single peak which revealed the presence of a single amplicon with specific amplification of the target (Fig. S4). Five genes (*SbZIP*2, 5, 6, 7, and 8) were expressed in root and shoot tissues of sorghum under both deficient and sufficient Zn supply (Fig. 7a, and b). In the shoot, eight genes (*SbZIP*2, 3, 4, 5, 6, 7, 8, and 10) out of 10 were expressed under both deficient and sufficient Zn supply (Fig. 7a). Among these, expressions of *SbZIP*2, 4, 5, 6, 7, 8, and 10 were significantly higher in shoot tissues under deficient Zn compared to sufficient Zn. *SbZIP*6 and 7 were upregulated fourfold, and *SbZIP*4, 8, and 10 were upregulated more than onefold under deficient Zn compared to sufficient Zn supply in shoot tissues (Fig. 7a). Expression levels of *SbZIP*2 and 5 were slightly higher in shoot tissues under deficient Zn compared to sufficient Zn. Expression levels of these five genes were significantly higher (> onefold) under

deficient Zn compared to sufficient Zn in the shoot tissue of sorghum.

Seven genes (*SbZIP*2, 5, 6, 7, 8, 9, and 10) were expressed in the root under deficient and sufficient Zn conditions (Fig. 7b). In the root, *SbZIP*4 was expressed under sufficient Zn and not under deficient Zn supply. *SbZIP*9 was expressed only in root and not in shoot tissues under deficient and sufficient Zn supply. *SbZIP*9 was strongly induced (fourfold) under deficient Zn in root tissues compared to sufficient Zn (Fig. 7b). *SbZIP*5 was expressed equally in root under deficient and sufficient Zn conditions. Expression of *SbZIP*6 was significantly higher (fivefold) in root under deficient Zn compared to sufficient Zn. In the root tissues, *SbZIP*7 and 8 were upregulated 2- and onefold under deficient Zn compared to sufficient Zn supply. *SbZIP*1 and 3 were not expressed in root under deficient and sufficient Zn conditions. The above results revealed that more *SbZIP* family genes were expressed in shoot and root tissues of sorghum under deficient Zn supply (Fig. 7a and b).

Discussion

Zn is one of the essential micro-nutrients required for plant growth and metabolism (Krishna et al. 2017). Zn deficiency is found in every part of the world and in almost all crops (Sadeghzadeh 2013). About 50% of the cereal cultivation land is Zn deficient worldwide (Vadlamudi et al. 2020). Zn deficiency reduces the biochemical and physiological processes of the plant (Marichali et al. 2014). Sufficient levels of Zn are essential for better plant growth and agricultural productivity (Rehman et al. 2018; Zulfiqar et al. 2020). Farmers use Zn fertilizer to improve

Fig. 4 Homology modeling of SbZIP proteins of sorghum. Homology modeling of SbZIP transporters was performed using *Bordetella bronchiseptica* ZIP (BbZIP) (PDB ID: 5TSA) as a template. The models consisting of the whole structure (a) and ligand-binding site (b) are shown for template BbZIP and sorghum transporters SbZIP1 to 10. The models for the truncated SbZIP proteins were generated using Modeller v9.22, which created 100 models for each target protein. The models were then ranked based on their modeller objective function value. The five best models were analyzed using MolProbity (<http://molprobity.biochem.duke.edu/>). Ramachandran plots were generated for these five models, and finally, the best model was chosen based on the % residues in the favored and allowed regions and the number of outliers. The BbZIP structure was visualized using LiteMol viewer (<https://www.litemol.org/Viewer/>)

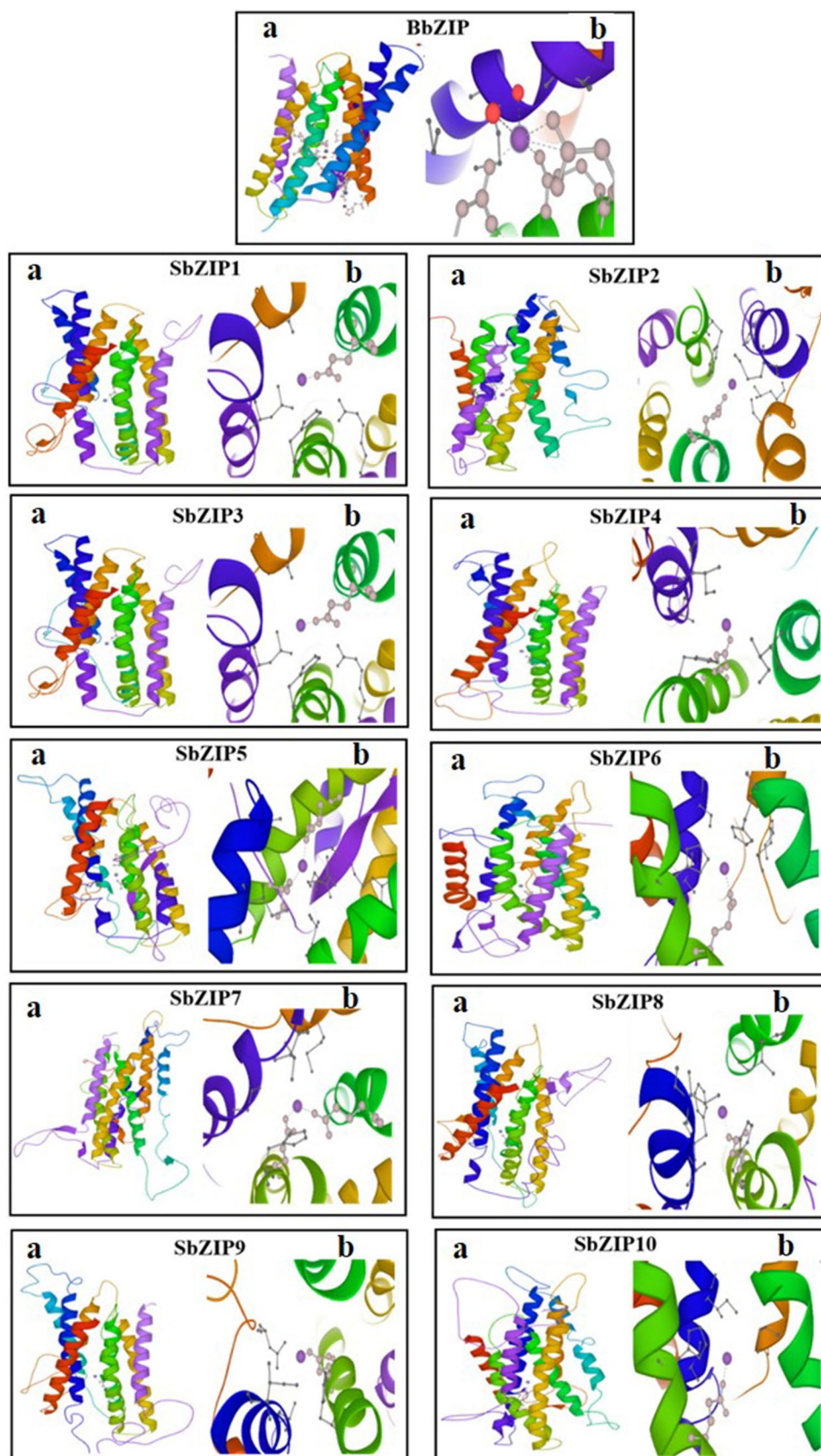
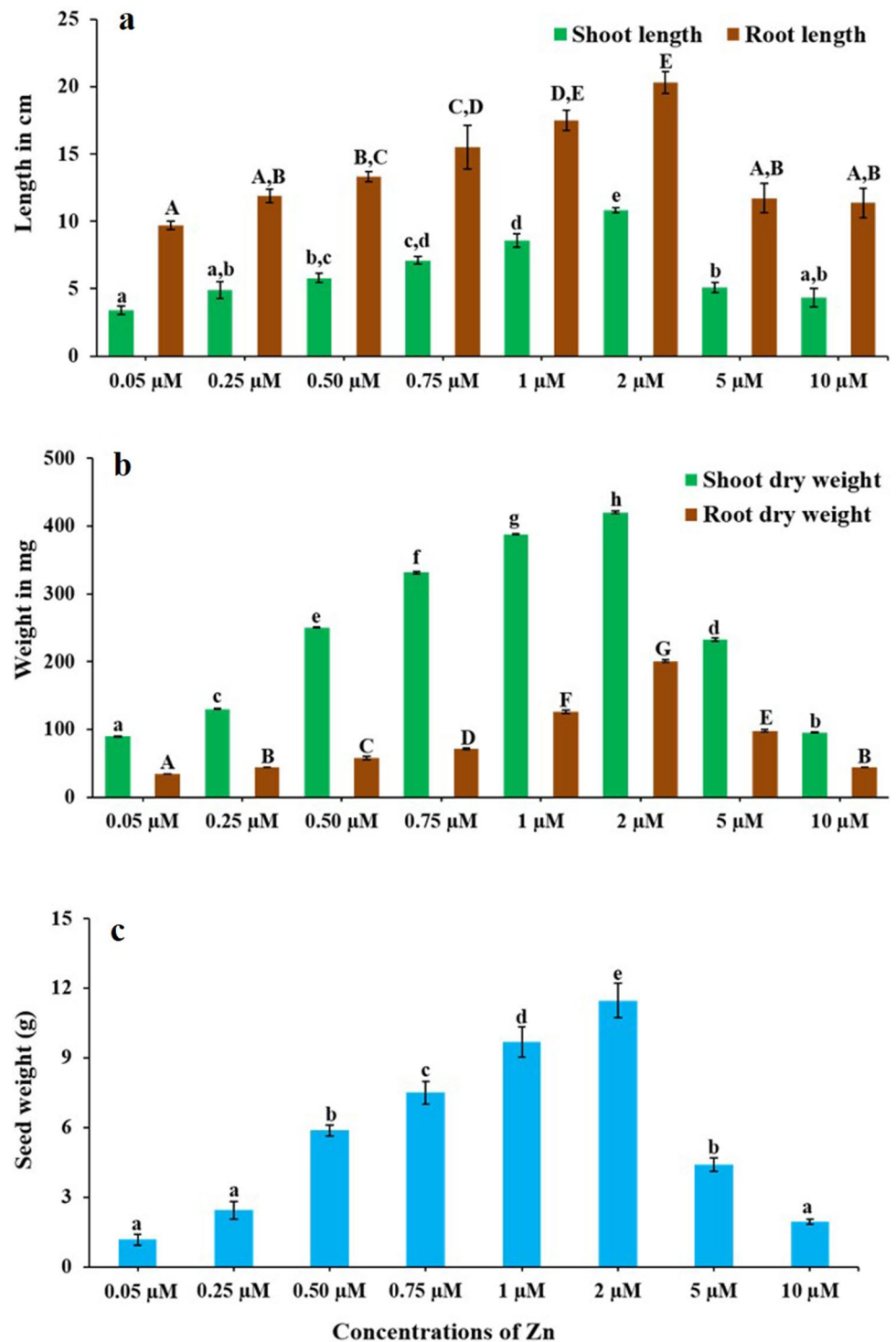


Fig. 5 Effect of Zn on the length and dry weight of shoot and root and seed weight of sorghum grown under various concentrations of Zn. Length (a) and dry weight (b) of shoot and root and seed weight (c) of sorghum grown under eight various (0.05, 0.25, 0.5, 0.75, 1.0, 2.0, 5.0, and 10 μ M) concentrations of Zn. The length and dry weight of the shoot and root were measured in 28-day-old plants under various concentrations of Zn. Seed weight was recorded after the maturation of seeds in sorghum. Data shown are means $\pm$ standard deviation (SD), $n=9$. All tissues produced significantly different responses under various levels of Zn ($P<0.05$) based on a Tukey post-hoc test



crop yields (Wasaya et al. 2017). In some developing countries, farmers apply too much Zn fertilizer to increase the yield of crops. Excessive use of Zn fertilizer also adversely affects crop development and yield (Vadlamudi et al. 2020). The price of Zn fertilizer is very high, so farmers in developing countries cannot afford to buy Zn fertilizer (Krishna et al. 2020). It is important to develop

new cultivars that grow well under deficient Zn, which will help improve poor farmers' lives. Sorghum is mostly cultivated in developing countries, and no clear report is available on the influence of Zn on the growth and yield of sorghum. However, some physiological and biochemical traits were analyzed in the BTx623 genotype of sorghum under Zn deficiency condition (Li et al. 2013a). The

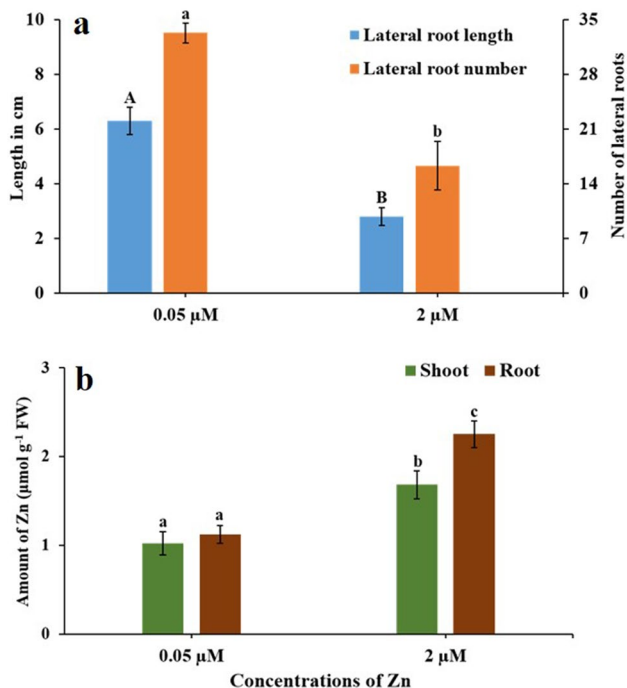


Fig. 6 Measurement of lateral root length (LRL) and lateral root number (LRN) and an assay of Zn contents in leaf, shoot, and root tissues of sorghum grown under deficient and sufficient Zn conditions. **a** The LRL and LRN were measured in 28-day-old plants of sorghum grown under deficient and sufficient Zn conditions. The LRL and LRN were measured using ImageJ scientific software. **b** Zn contents were analyzed in leaf, shoot and root tissues of 28-day-old plants of sorghum grown under deficient and sufficient Zn conditions. Data shown are means $\pm$ SD, $n=9$. Values followed by the same letter are not significantly different ($P>0.05$) based on a Tukey post-hoc test

complete annotated genome sequence is also available for this genotype. In general, BTx623 has short stature and an early maturing genotype. However, it is susceptible to sugarcane aphids and sensitive to low nitrogen (Cooper et al. 2019). Therefore, it is often used in functional comparative studies.

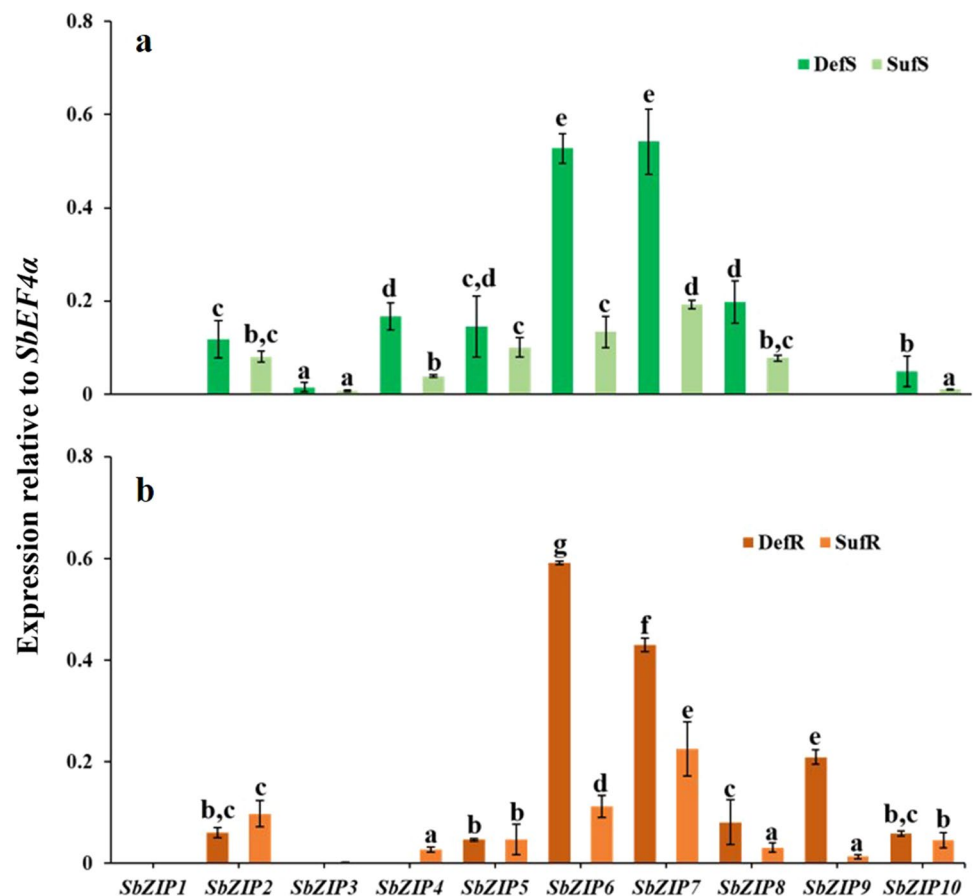
In the present study, we have identified Zn's optimum, sufficient, and toxicity levels for the APK-1 genotype of sorghum growth under greenhouse conditions. Like BTx623, APK 1 is a medium-duration and high-yielding genotype (Seetharam and Ganesamurthy 2013). It is tolerant of growing under certain diseases. Therefore, we selected the APK-1 genotype for our study. We found that 2 μM Zn was sufficient (optimum) for sorghum growth and development based on the phenotypic traits, Zn content, and yield data (Figs. 5 and 6). Li et al. (2013a) also reported that 2 μM Zn is sufficient for sorghum growth and development based on their phenotypic traits and Zn content data. A range of Zn concentration for normal growth was reported in many other crop plants like wheat (1 μM), rice (1.5 μM), barley (0.5 μM), and mungbean (2.0 μM) (Cakmak et al. 1996;

Impa et al. 2013). In this study, Zn deficiency reduced the SL and PRL by 68.7 and 51.5%, respectively (Fig. 5a). The present study observed a similar reduction in SDW and RDW of sorghum grown under the same deficient Zn supply. More than 75% of the shoot and root dry weight was affected by Zn deficiency (Fig. 5b). In sorghum, the dry weight of leaf and root was reduced by 43 and 33%, respectively, under deficient Zn compared to sufficient Zn (Li et al. 2013a). In rice, the plant height, PRL, SDW, RDW, and total dry matter were reduced ($>30\%$) under deficient Zn (0.005 μM) compared with the sufficient Zn (1.5 μM) (Impa et al. 2013). Zn plays a key role in enzyme activity. It is the only element necessary to activate all six classes of enzymes (Sadeghzadeh 2013). Zn is also a co-factor for around 300 individual plant enzymes (Rathore and Mohit 2013). Plant growth and development are affected when specific enzymes are active in plant tissues. Hence, the Zn deficiency severely affects plants' growth and biomass.

During nutrient (including Zn) deficiency, phytohormones and sugars are involved in reducing PRL and improving the LRL and LRN (Maharajan et al. 2018). In this study, sorghum produced higher LRN under deficient Zn conditions (Fig. 6a). The LRL also increased in sorghum under deficient Zn compared to the sufficient Zn supply (Fig. 6a). A similar result was also found in sorghum under Zn deficient condition (Li et al. 2013a). The lateral root initiation and elongation play an important role in the improved uptake of Zn from the soil under deficient Zn supply. The LRL and LRN were increased in barley under deficient Zn (0 μM) compared to sufficient Zn (0.5 μM) conditions (Genc et al. 2007). All these results indicated that sorghum grown under deficient Zn produced more LRN and higher LRL. Zn is an essential plant micronutrient, but excess Zn can be harmful (Wang et al. 2009). We also noticed through this study that excess Zn ($>5 \mu\text{M}$) reduced the phenotypic traits and yield in sorghum (Fig. 5a–c). Excess Zn reduces the uptake of other macro- and micro-nutrients such as P, copper (Cu), Fe, manganese (Mn), and magnesium (Mg) (Wang et al. 2009; Vadlamudi et al. 2020). It has been confirmed that excess Zn reduces the concentrations of P, Cu, Fe, Mn, and Mg in rapeseed leaf and root tissues (Vadlamudi et al. 2020). In our study, excess Zn also influences the phenotypic and yield traits of sorghum, which might be due to the inhibition of the uptake of other nutrients. The toxicity concentration of Zn for sorghum has not yet been identified. Therefore, this study may help researchers to know the exact concentration of Zn for sorghum growth and yield.

We observed deficient Zn supply reduced sorghum's flower quality and SW (Fig. 1c and 5c). In this study, the SW of sorghum was reduced by more than 80% under deficient Zn compared to sufficient Zn supply. Many reports are also available on Zn deficiency-induced reductions in the yield and quality of flowers for crops like rice (Chen

Fig. 7 Quantitative real-time RT-PCR (qRT-PCR) analysis of 10 *SbZIP* genes in the shoot, and root tissues of sorghum grown under deficient and sufficient Zn conditions. Expression levels of *SbZIP* genes in the shoot (a) and root (b) samples of 28-day-old plants. *SbEF4a* was chosen as a reference gene for the normalization of gene expression. The Cycle threshold (Ct) values of all tissues of sorghum were analyzed using the formula $2^{-\Delta\Delta Ct}$. Three biological replicates, each having three technical replicates ($n=9$) of the shoot, leaf, and root sample, were used for qRT-PCR analysis. DefS and DefR are represented as deficient shoot and root, respectively (this indicates that shoot and root tissues of sorghum were collected under deficient Zn condition). SufS, and SufR are represented as sufficient shoot and root, respectively (this indicates that shoot and root tissues of sorghum were collected under sufficient conditions)



et al. 2008), wheat (Zulfiqar et al. 2020), maize (Wang and Jin 2005), and barley (Genc et al. 2004). It clearly shows that a sufficient concentration of Zn is needed for crop plants' better growth and yield, including sorghum. In this study, we have successfully determined the sufficient and deficient concentrations of Zn for sorghum growth and yield in greenhouse conditions. This study may lay the foundation for further studies to help improve sorghum production under low Zn soils.

Plant ZIP family transporters are primarily involved in Zn uptake, transport, and distribution (Krishna et al. 2020). However, ZIP family transporters are also reported to be involved in the transport of other divalent metal cations such as Mn^{2+} , Fe^{2+} , cadmium (Cd^{2+}), cobalt (Co^{2+}), Cu^{2+} and nickel (Ni^{2+}) in addition to Zn (Pedas et al. 2009). ZIP transporters play a crucial role in the bio-fortification of grains with Zn (Singh and Prasanna 2020). The ZIP family transporters have been identified, and their expression profiles were analyzed in various tissues of rice, maize, wheat, barley, and foxtail millet (Krishna et al. 2020).

Structural characterization of Zn transporters in plant species, including sorghum, is still limited. The high-resolution crystal structure of BbZIP is available now, and its functional residues are well characterized (Zhang et al.

2017). Recently, the functional residues of plant ZIP proteins were predicted by BbZIP (Krishna et al. 2020). For multiple sequence alignment and homology modeling, we used protein sequences of BbZIP and truncated protein sequences of SbZIP to identify the conserved residues involved in Zn binding and transport (Figs. 3 and 4). The most important functional residues of BbZIP are found in sorghum SbZIPs. The functional residue of Asp144 is involved in metal release in BbZIP. The ASP144 is conserved in SbZIP7. Two residues (His177 and Gly182) are involved in the metal release from the metal-binding site (Zhang et al. 2017). These two residues are fully conserved in all SbZIP family transporters (Fig. 3; Table 3). Leu200, Glu211, and Gly212 are essential for metal-binding (Zhang et al. 2017), and these all are fully conserved in all SbZIPs of sorghum except for SbZIP1 and 2. However, based on the homology modeling, it has been observed that most of the Zn^{2+} binding site residues are not conserved in SbZIP transporters. Glu (Glu 181) is the only residue conserved at the Zn^{2+} binding of all the SbZIP proteins (Table 4; Fig. 4). Apart from His177, which is most prevalent among the SbZIPs, other residues are not conserved, and SbZIPs have many novel residues like Arg,

Leu, Ala, and Ser at the ligand-binding site. Plant ZIP proteins, including SbZIPs, may have slight overlapping but distinct mechanisms of Zn transport (Krishna et al. 2020). This calls for further functional studies of plant ZIP transporters by site-directed mutagenesis of functional residues and nutrient uptake assays in yeast or *in planta* to characterize the functional residues.

The phylogeny analysis of SbZIPs with other plants showed that SbZIP2 is closely clustered with SiZIP3 of foxtail millet. SiZIP3 was highly expressed in the leaf, root, stem, and spica tissues of foxtail millet grown under drought stress (Alagarasan et al. 2017). In our study, SbZIP2 was highly expressed in leaf and shoot tissues under deficient Zn. The SbZIP2 transporter might be active in various tissues for efficient Zn transport in sorghum. We need to analyze the expression pattern with functional characterization of SbZIP2 in various sorghum tissues under drought stress, which may help identify the role of SbZIP2 in sorghum under drought stress. Four SbZIP proteins (SbZIP4, 5, 6, and 8) are closely clustered with ZmZIP5, ZmZIP9, ZmZIP11, and SvZIP8, respectively. ZmZIP5, 9, and 11 were highly expressed in flag leaf tissue of maize under deficient Zn conditions (Mondal et al. 2014). Through a qRT-PCR study, we found that all these genes (SbZIP4, 5, 6, and 8) were highly expressed in the shoot tissue of sorghum under deficient Zn. Three genes (SbZIP4, 6, and 8) were highly expressed in the sorghum leaf under deficient Zn.

Further functional characterization of SbZIP4, 5 6 and 8 may help to decipher its role in the mobilization of Zn from flag leaf to developing spikes for the accumulation of a large amount of Zn in the developing spike. Some SbZIP transporters are distantly clustered with green foxtail and foxtail millet. No ZIP family transporters have yet been characterized in these two millets under deficient Zn conditions. The construction of a phylogenetic tree may help to understand the role of SvZIP and SiZIP in these two millets.

ZIP family transporters have not yet been identified in sorghum. However, the expression patterns of miRNAs were analyzed in leaf and root tissues of sorghum under deficient and sufficient conditions (Li et al. 2013a). They have reported that miRNAs are essential regulators of the response of Zn deficiency in sorghum. We have identified the SbZIP family transporters and analyzed their expression patterns in shoot and root tissues of sorghum under deficient and sufficient Zn conditions for the first time through this study. The ZIP family transporter has already been proven to be involved in Zn²⁺ transport under Zn deficiency conditions in monocot (rice) (Huang et al. 2020). Hence, this study will help to improve Zn uptake under deficient Zn conditions. Out of 10 ZIPs, five genes

(SbZIP2, 5, 6, 7 and 8) were expressed in shoot and root tissues of sorghum grown under both deficient and sufficient Zn supply (Fig. 7a, b). SbZIP4, 6, 7, 8, and 10 were upregulated in shoot under deficient Zn compared to the sufficient Zn supply (Fig. 7a). In rice, OsZIP4 and 8 were upregulated in shoot tissues under deficient Zn supply, and they have been reported to be involved in the translocation of Zn from root to shoot (Ishimaru et al. 2005; Ji et al. 2022). TaZIP6 and 7 were highly expressed in shoot tissues of wheat under deficient Zn, which may play a key role in the translocation of Zn from root to shoot tissues (Evens et al. 2017). Based on the results, we assume that SbZIP4, 6, 7, and 8 may be involved in the translocation of Zn from root to shoot under deficient Zn conditions. More SbZIP genes were upregulated in root tissues of sorghum under deficient Zn conditions (Fig. 7b). In the present study, SbZIP9 was expressed only in the root of sorghum under deficient and sufficient Zn supply. SbZIP9 was also upregulated under deficient Zn compared to sufficient Zn supply. This is very interesting, because this was not detected in the shoot tissues of sorghum under either Zn conditions. SbZIP6, 7, 8, and 10 were upregulated under deficient Zn compared to the sufficient Zn in the root tissue of sorghum. OsZIP7 and 8 were highly expressed in rice, and they have been reported to be responsible for Zn uptake from soil under a deficient Zn supply (Ishimaru et al. 2005). Three (ZmZIP6, 7, and 8) and two (TaZIP6 and 7) genes were upregulated in root tissues of maize and wheat, respectively, under deficient Zn supply (Li et al. 2013b; Mondal et al. 2014; Evens et al. 2017). All these genes are reported responsible for Zn uptake from the soil. The upregulated genes (SbZIP6, 7, 8, and 10) in the root tissue of sorghum may help to uptake Zn from soil under deficient Zn. In the present study, the SbZIP1 was not expressed in any tissues under deficient and sufficient Zn conditions. It might be expressed in other tissues like a leaf, stem, flag leaves, flowers, etc., under Zn deficiency and sufficiency conditions that need further investigation. This study might be a valuable basis for future studies aiming to further characterize these transporters in sorghum at structural and functional levels. Further functional characterization of SbZIP genes will help to understand each gene's role and improve the growth and development of sorghum under deficient Zn. The functional characterization of SbZIP family transporters in sorghum may also help to improve the Zn-use efficiency of sorghum and other closely related cereals, particularly small millets.

Conclusion

Increasing sorghum production can alleviate food shortages among poor people worldwide, particularly in developing countries. Improving the sorghum yield under low-nutrient soil may help to strengthen food security. In the present study, we have identified the sufficient and toxic level of Zn based on the phenotypic, Zn content, and yield traits for sorghum growth and development. This study may help conduct further physiological, phenomic, and molecular experiments in sorghum to improve Zn uptake and fortification. More *SbZIP* genes were upregulated in shoot and root tissues of sorghum under deficient Zn compared to sufficient Zn.

Further study is required to identify the genes responsible for Zn uptake, translocation, and remobilization. The clustered, regularly interspaced, short palindromic repeats (CRISPR)/CRISPR-associated protein 9 (Cas9) system has emerged as a most popular tool for precisely editing the genomes of higher plants. This tool can target a sequence for gene knock-in, knock-out, and replacement and monitor and regulate gene expression (Ceasar et al. 2022; Krishna et al. 2022b; Maharajan et al. 2022c). Development of sorghum mutants through the CRISPR/Cas9 system (knock-in of the shoot and root-specific gene at the specific site of the genome) will help enhance the shoot and root growth of sorghum under deficient Zn.

Author contribution statement TM, TPAK, SI, and SAC conceived and designed the experiments. TM, TPAK, and SAC performed the experiments. TM, TPAK, and SAC analyzed the data. TM and TPAK wrote the manuscript. SAC and SI assisted, edited, and updated the manuscript. SAC contributed critically to revising and improving the manuscript for publication.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00425-023-04076-5>.

Acknowledgements The authors thank the Tamil Nadu Agricultural University, Coimbatore, India, for providing sorghum seed.

Funding This work was funded by Rajagiri College of Social Sciences, Kochi, Kerala, India, Under Seed Money for Faculty Minor Research.

Data availability All data generated or analyzed during this study are included in this article.

Declarations

Conflict of interest The authors declare that they have no competing interests.

References

- Abah CR, Ishiwu CN, Obiegbuna JE, Oladejo AA (2020) Sorghum grains: nutritional composition, functional properties and its food applications. *Eur J Nutr Food Saf* 12:101–111
- Alagarasan G, Dubey M, Aswathy KS, Chandel G (2017) Genome wide identification of orthologous *ZIP* genes associated with zinc and iron translocation in *Setaria italica*. *Front Plant Sci* 8:775
- Amini S, Arsova B, Hanikenne M (2022) The molecular basis of zinc homeostasis in cereals. *Plant Cell Environ* 45:1339–1361
- Bashir K, Ishimaru Y, Nishizawa NK (2012) Molecular mechanisms of zinc uptake and translocation in rice. *Plant Soil* 361:189–201
- Cakmak I, Sari NE, Marschner H, Kalayci M, Yilmaz A, Eker SE, Gulut KY (1996) Dry matter production and distribution of zinc in bread and durum wheat genotypes differing in zinc efficiency. *Plant Soil* 180:173–181
- Ceasar SA, Hodge A, Baker A, Baldwin SA (2014) Phosphate concentration and arbuscular mycorrhizal colonisation influence the growth, yield and expression of twelve *PHT1* family phosphate transporters in foxtail millet (*Setaria italica*). *PLoS ONE* 9:e108459
- Ceasar SA, Maharajan T, Hillary E, Krishna TPA (2022) Insights to improve the plant nutrient transport by CRISPR/Cas system. *Biotechn Adv* 59:107963
- Chen WR, Feng Y, Chao YE (2008) Genomic analysis and expression pattern of *OsZIP1*, *OsZIP3*, and *OsZIP4* in two rice (*Oryza sativa* L.) genotypes with different zinc efficiency. *Russ J Plant Physiol* 55:400–409
- Clemens S (2001) Molecular mechanisms of plant metal tolerance and homeostasis. *Planta* 212:475–486
- Conte SS, Walker EL (2011) Transporters contributing to iron trafficking in plants. *Mol Plant* 4:464–476
- Cooper EA, Brenton ZW, Flinn BS, Jenkins J, Shu S, Flowers D, Kresovich S (2019) A new reference genome for *Sorghum bicolor* reveals high levels of sequence similarity between sweet and grain genotypes: implications for the genetics of sugar metabolism. *BMC Genomics* 20:420
- Eide D, Broderius M, Fett J, Lou GM (1996) A novel iron-regulated metal transporter from plants identified by functional expression in yeast. *Proc Natl Acad Sci USA* 93:5624–5628
- Evens NP, Buchner P, Williams LE, Hawkesford MJ (2017) The role of ZIP transporters and group F bZIP transcription factors in the Zn-deficiency response of wheat (*Triticum aestivum*). *Plant J* 92:291–304
- Genc Y, McDonald GK, Graham RD (2004) Differential expression of zinc efficiency during the growing season of barley. *Plant Soil* 263:273–282
- Genc Y, Huang CY, Langridge P (2007) A study of the role of root morphological traits in growth of barley in zinc-deficient soil. *J Exp Bot* 58:2775–2784
- Gerrano AS, Labuschagne MT, van Biljon A, Shargie NG (2014) Genetic variability among sorghum accessions for seed starch and stalk total sugar content. *Sci Agric* 71:472–479
- Gupta N, Ram H, Kumar B (2016) Mechanism of zinc absorption in plants: uptake, transport, translocation and accumulation. *Rev Environ Sci Biotechnol* 15:89–109
- Hall JL, Williams LE (2003) Transition metal transporters in plants. *J Exp Bot* 54:2601–2613
- Huang S, Sasaki A, Yamaji N, Okada H, Mitani-Ueno N, Ma JF (2020) The ZIP transporter family member OsZIP9 contributes to root zinc uptake in rice under zinc-limited conditions. *Plant Physiol* 183:1224–1234

- Impa SM, Gramlich A, Tandy S, Schulin R, Frossard E, Johnson-Beebout SE (2013) Internal Zn allocation influences Zn deficiency tolerance and grain Zn loading in rice (*Oryza sativa* L.). *Front Plant Sci* 4:534
- Ishimaru Y, Suzuki M, Kobayashi T, Takahashi M, Nakanishi H, Mori S, Nishizawa NK (2005) OsZIP4, a novel zinc-regulated zinc transporter in rice. *J Exp Bot* 56:3207–3214
- Ishimaru Y, Masuda H, Suzuki M, Bashir K, Takahashi M, Nakanishi H, Mori S, Nishizawa NK (2007) Overexpression of the OsZIP4 zinc transporter confers disarrangement of zinc distribution in rice plants. *J Exp Bot* 58:2909–2915
- Ji C, Li J, Jiang C, Zhang L, Shi L, Xu F, Cai H (2022) Zinc and nitrogen synergistic act on root-to-shoot translocation and preferential distribution in rice. *J Adv Res* 35:187–198
- Kolaj-Robin O, Russell D, Hayes KA, Pembroke JT, Soulimane T (2015) Cation diffusion facilitator family: structure and function. *FEBS Lett* 589:1283–1295
- Krishna TPA, Ceasar SA, Maharajan T, Ramakrishnan M, Duraipandiyan V, Al-Dhabi NA, Ignacimuthu S (2017) Improving the zinc-use efficiency in plants: a review. *Sabara J Breed Genet* 49:211–230
- Krishna TPA, Maharajan T, Roch GV, Ignacimuthu S, Ceasar SA (2020) Structure, function, regulation and phylogenetic relationship of ZIP family transporters of plants. *Front Plant Sci* 11:662
- Krishna TPA, Maharajan T, Ceasar SA (2022a) The role of membrane transporters in the biofortification of zinc and iron in plants. *Biol Trace Elem Res*. <https://doi.org/10.1007/s12011-022-03159-w>
- Krishna TPA, Maharajan T, Ceasar SA (2022b) Application of CRISPR/Cas9 genome editing system to reduce the pre-and post-harvest yield losses in cereals. *Open Biotech J* 16:1–9
- Lee S, Kim SA, Lee J, Guerinot ML, An G (2010) Zinc deficiency-inducible *OsZIP8* encodes a plasma membrane-localized zinc transporter in rice. *Mol Cells* 29:551–558
- Li Y, Zhang Y, Shi D, Liu X, Qin J, Ge Q, Xu L, Pan X, Li W, Zhu Y, Xu J (2013a) Spatial-temporal analysis of zinc homeostasis reveals the response mechanisms to acute zinc deficiency in *Sorghum bicolor*. *New Phytol* 200:1102–1115
- Li S, Zhou X, Huang Y, Zhu L, Zhang S, Zhao Y, Guo J, Chen J, Chen R (2013b) Identification and characterization of the zinc-regulated transporters, iron-regulated transporter-like protein (ZIP) gene family in maize. *BMC Plant Biol* 13:114
- Liu XB, Yang Q, Chu TD, Wang SH, Li SR, Wu XF (1993) Effect of zinc application on corn. *Acta Pedofol Sin* 30:153–162
- Macnair MR, Smirnoff N (1999) Use of zinc to study uptake and accumulation of zinc by zinc tolerant and hyperaccumulating plants. *Commun Soil Sci Plant Anal* 30:1127–1136
- Maharajan T, Ceasar SA, Krishna TPA, Ramakrishnan M, Duraipandiyan V, Naif Abdulla AD, Ignacimuthu S (2018) Utilization of molecular markers for improving the phosphorus efficiency in crop plants. *Plant Breed* 137:10–26
- Maharajan T, Ceasar SA, Krishna TPA, Ignacimuthu S (2019) Phosphate supply influenced the growth, yield and expression of *PHT1* family phosphate transporters in seven millets. *Planta* 250:1433–1448
- Maharajan T, Krishna TP, Kiriyanthan RM, Ignacimuthu S, Ceasar SA (2021a) Improving abiotic stress tolerance in sorghum: focus on the nutrient transporters and marker-assisted breeding. *Planta* 254:90
- Maharajan T, Ceasar SA, Krishna TPA, Ignacimuthu S (2021b) Finger millet [*Eleusine coracana* (L.) Gaertn]: an orphan crop with a potential to alleviate the calcium deficiency in the semi-arid tropics of Asia and Africa. *Front Sustain Food Syst* 5:684447
- Maharajan T, Ceasar SA, Krishna TPA (2022a) Finger Millet (*Eleusine coracana* (L.) Gaertn): nutritional importance and nutrient transporters. *Crit Rev Plant Sci* 41:1–31
- Maharajan T, Chellasamy G, Krishna TPA, Ceasar SA, Yun K (2022b) The role of metal transporters in phytoremediation: a closer look at *Arabidopsis*. *Chem* 310:136881
- Maharajan T, Krishna TPA, Rakkammal K, Ceasar SA, Ramesh M (2022c) Application of CRISPR/Cas system in cereal improvement for biotic and abiotic stress tolerance. *Planta* 256:106
- Marichali A, Dallali S, Ouerghemmi S, Sebei H, Hosni K (2014) Germination, morpho-physiological and biochemical responses of coriander (*Coriandrum sativum* L.) to zinc excess. *Ind Crops Prod* 55:248–257
- Mondal TK, Ganie SA, Rana MK, Sharma TR (2014) Genome-wide analysis of zinc transporter genes of maize (*Zea mays*). *Plant Mol Biol Report* 32:605–616
- Mundia CW, Secchi S, Akamani K, Wang G (2019) A regional comparison of factors affecting global sorghum production: the case of North America, Asia and Africa's Sahel. *Sustainability* 11:2135
- Nozoye T, Nagasaka S, Kobayashi T, Sato Y, Uozumi N, Nakanishi H, Nishizawa NK (2015) The phytosiderophore efflux transporter TOM2 is involved in metal transport in rice. *J Biol Chem* 290:27688–27699
- Paterson AH, Bowers JE, Bruggmann R, Dubchak I, Grimwood J, Gundlach H, Haberer G, Hellsten U, Mitros T, Poliakov A, Schmutz J (2009) The *Sorghum bicolor* genome and the diversification of grasses. *Nature* 457:551–556
- Pedraza P, Schjoerring JK, Husted S (2009) Identification and characterization of zinc-starvation-induced ZIP transporters from barley roots. *Plant Physiol Biochem* 47:377–383
- Ramesh SA, Shin R, Eide DJ, Schachtman DP (2003) Differential metal selectivity and gene expression of two zinc transporters from rice. *Plant Physiol* 133:126–134
- Rathore JS, Mohit U (2013) Investigation of zinc concentration in some medicinal plant leaves. *Res J Pharm Sci* 2:15–17
- Reddy PS, Srinivas Reddy D, Sivasakthi K, Bhatnagar-Mathur P, Vadez V, Sharma KK (2016) Evaluation of sorghum [*Sorghum bicolor* (L.)] reference genes in various tissues and under abiotic stress conditions for quantitative real-time PCR data normalization. *Front Plant Sci* 7:529
- Rehman A, Farooq M, Ozturk L, Asif M, Siddique KH (2018) Zinc nutrition in wheat-based cropping systems. *Plant Soil* 422:283–315
- Roch GV, Maharajan T, Krishna TP, Ignacimuthu S, Ceasar SA (2020) Expression of *PHT1* family transporter genes contributes for low phosphate stress tolerance in foxtail millet (*Setaria italica*) genotypes. *Planta* 252:98
- Sadeghzadeh B (2013) A review of zinc nutrition and plant breeding. *J Soil Sci Plant Nutr* 13:905–927
- Seetharam K, Ganesamurthy K (2013) Characterization of sorghum genotypes for yield and other agronomic traits through genetic variability and diversity analysis. *Elect J Plant Breed* 4:1073–1079
- Singh D, Prasanna R (2020) Potential of microbes in the biofortification of Zn and Fe in dietary food grains. A review. *Agron Sustain Dev* 40:1–21
- Suzuki M, Takahashi M, Tsukamoto T, Watanabe S, Matsuhashi S, Yazaki J, Kishimoto N, Kikuchi S, Nakanishi H, Mori S, Nishizawa NK (2006) Biosynthesis and secretion of mugineic acid family phytosiderophores in zinc-deficient barley. *Plant J* 48:85–97
- Tiong J, McDonald G, Genc Y, Shirley N, Langridge P, Huang CY (2015) Increased expression of six ZIP family genes by zinc (Zn) deficiency is associated with enhanced uptake and

- root-to-shoot translocation of Zn in barley (*Hordeum vulgare*). *New Phytol* 207:1097–1109
- Vadlamudi K, Upadhyay H, Singh A, Reddy M (2020) Influence of zinc application in plant growth: an Overview. *Eur J Mol Clin Med* 7:2321–2327
- Von Wiren N, Marschner H, Romheld V (1996) Roots of iron-efficient maize also absorb phytosiderophore-chelated zinc. *Plant Physiol* 111:1119–1125
- Wang H, Jin JY (2005) Photosynthetic rate, chlorophyll fluorescence parameters, and lipid peroxidation of maize leaves as affected by zinc deficiency. *Photosynthetica* 43:591–596
- Wang C, Zhang SH, Wang PF, Hou J, Zhang WJ, Li W, Lin ZP (2009) The effect of excess Zn on mineral nutrition and antioxidative response in rapeseed seedlings. *Chemosphere* 75:1468–1476
- Wang J, Yang Y, Liao L, Xu J, Liang X, Liu W (2019) Genome-wide identification and functional characterization of the phosphate transporter gene family in *Sorghum*. *Biomolecules* 9:670
- Wasaya A, Shahzad Shabir M, Hussain M, Ansar M, Aziz A, Hassan W, Ahmad I (2017) Foliar application of zinc and boron improved the productivity and net returns of maize grown under rainfed conditions of Pothwar plateau. *J Soil Sci Plant Nutr* 17:33–45
- Yuvaraj M, Subramanian KS (2020) Significance of zinc in plant nutrition. *Biot Res Today* 2:823–825
- Zhang T, Liu J, Fellner M, Zhang C, Sui D, Hu J (2017) Crystal structures of a ZIP zinc transporter reveal a binuclear metal center in the transport pathway. *Sci Adv* 3:e1700344
- Zulfiqar U, Hussain S, Ishfaq M, Matloob A, Ali N, Ahmad M, Alyemeni MN, Ahmad P (2020) Zinc-induced effects on productivity, zinc use efficiency, and grain biofortification of bread wheat under different tillage permutations. *Agronomy* 10:1566

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.